

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I Sanjanaa Sree.P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



$20 + 28 + 20 + 20 + 8$

$= 96$



Unit 2: Assessment Tool 3

Dry Run Evaluation

Course Code: CSA1710 Artificial
Intelligence

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1). Online Search Agent:

Problem Statement:

A Robot is placed in an unfamiliar building.

Initially it knows only:

(*) $A \rightarrow B$

(*) $A \rightarrow C$

Later, while exploring, it discovers:

(*) $B \rightarrow D$

(*) $D \rightarrow G$ (Goal)

Online Search Concept:

An online search agent does not know the complete environment beforehand. It discovers new states only when it moves through the environment. The agent updates its internal map as it explores.

Initial knowledge :

State

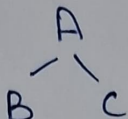
known connections

A

B, C

current Position = A .

known Map :



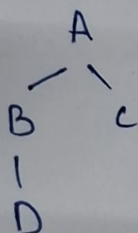
Goal G is not yet known .

Step 1: Start at A

The Robot is at State A .

Available moves : B and C . Choose B .

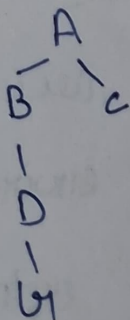
Path : $A \rightarrow B$ $B \rightarrow D$.



Step 2: Explore D .

From B, the robot moves to D .

Path : $A \rightarrow B \rightarrow D$. $D \rightarrow G$



Step 3: Reach Goal

Robot moves from D to G .

Path : $A \rightarrow B \rightarrow D \rightarrow G$.

1. Constraint Satisfaction Problem (CSP):

Problem Statement:

Four Students A, B, C and D must be assigned to one of the following presentation slots:

(*) 9 AM

(*) 10 AM

(*) 11 AM

Constraints:

(*) $A \neq 9 \text{ AM}$

(*) B before C

(*) D not at the same time as A.

(*) $C \neq 11 \text{ AM}$

(i) Variables:

The variables are: A, B, C, D.

Each variable represents the presentation time assigned to a student.

(ii) Domains:

Each student can initially take any slot.

Student

Domain

A

$\{9, 10, 11\}$

B

$\{9, 10, 11\}$

C

$\{9, 10, 11\}$

D

$\{9, 10, 11\}$

(iiB) Constraints:

Student	Constraint
A	Not 9 AM
B	Must be before C
C	Not 11 AM
D	Cannot be same time as A

Constraint Reduction:

For A : $A \neq 9$ $A = \{10, 11\}$

For C : $C \neq 11$ $C = \{9, 10\}$

Apply B before C :

Possible pairs : $B = 9, C = 10$

only valid ordering.

Thus, $B = 9$ and $C = 10$.

Choose A and A can be 11 AM.

Choose D : D cannot be same as A.

Since $A = 11$, $D = 10$ is valid.

Valid Assignment :

Student	Assigned slot
A	11 AM
B	9 AM
C	10 AM
D	10 AM

3). Map Colouring (SP using Back Tracking :

Problem Statement :

Regions A, B, C and D form a cycle :

(*) A adjacent to B and C

(*) B adjacent to A and D

(*) C adjacent to A and D

(*) D adjacent to B and C

Available colour : Red, Green, Blue.

Variable : A, B, C, D

Domain :

Region

A

B

C

D

Available colours

R, G, B

R, G, B

R, G, B

R, G, B

Constraints :

Adjacent regions must have different colours.

Region

A

B

C

D

Adjacent Regions

B, C

A, D

A, D

B, C

Backtracking :

Step 1 : Assign A

Choose A = Red.

Region	Color
A	Red

Step 2 : Assign B

B cannot be red because it is adjacent to A.

Choose : B = Green

Region	Color
A	Red
B	Green

Step 3 : Assign C

C is adjacent to A. Therefore C cannot be Red.

Choose C = Blue

Step 4 : Assign D

D is adjacent to B (Green) and C (Blue)

Hence D cannot be Green or Blue.

Choose D = Red.

Assignment :

Region	Color
A	Red
B	Green
C	Blue
D	Red